Ishan Shah

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Senior Engineering Leader | Test Automation Platforms, Quality Engineering & CI/CD | AI-Driven Test Tooling & Regulated Data Integrity

SUMMARY

Builder-leader with 15+ years across test automation, quality engineering, CI/CD, and platform/DevOps — the kind who stands up the platforms other teams adopt by default. Architected a Robot Framework test-automation platform (integration, E2E, regression, and performance suites with pipeline gating) for industrial control systems, and built TestBOT, a Gemini agent that auto-generates test cases from specs — cutting manual regression from days to <12 hours. Deep CI/CD expertise at scale: GitHub Actions, Jenkins, and Bazel monorepo build/test orchestration with gating and hermetic, ephemeral environments. Make build-vs-buy calls on TCO and signal quality, run quality as metrics (coverage, reliability/SLOs, DORA), and bring data-integrity and audit-trail rigor from FDA-regulated medical-device software and ISO 27001 / SOC 2 ownership. Lead small, high-leverage teams and scale through tooling and influence — hands-on in Python and TypeScript.

EXPERIENCE

CruxOCM Inc. — Head of Automation (reporting to the CTO) · Feb 2025 – Present

- **Test automation platform:** Architected and built a Robot Framework test-automation platform — integration, E2E, and regression suites with pipeline gating, fixtures, and reporting — adopted as the org default for validating OPC UA industrial control systems; cut manual regression from days to <12 hours.
- **AI-driven test generation:** Built TestBOT, a Gemini agent (`google-genai`) that auto-generates Robot Framework test cases from specs — applying LLMs to the test-authoring lifecycle.
- **CI/CD & test execution:** Owned CI/CD across GitHub Actions, Jenkins, Bazel (monorepo, hermetic builds), and Gradle with pipeline gating and self-service ephemeral environments; standardized Kubernetes control planes across GCP and Azure at 99.99% uptime with observability and SLOs.
- **Quality metrics & build-vs-buy:** Ran quality as first-class metrics (coverage, reliability/SLOs, DORA) with dashboards from PR to executive review; made build-vs-buy and vendor calls on AI and platform tooling by TCO and signal quality.
- **AI & agent platform:** Architected miniBOT, a Gemini multi-agent assistant (FastAPI/Cloud Run) grounded by RAG over BigQuery embeddings on Vertex AI, and shipped MCP servers (a published open-source Lever MCP server + Slack-native) — the AI/agent stack that a test-and-eval discipline governs.
- **Team, leadership & compliance:** Lead a small, high-leverage technical team — hire, mentor, and coach; scale other teams through tooling and influence; partner with the CEO/CTO on roadmap and OKRs; own ISO 27001 and SOC 2 (audit ownership) in a safety-critical, regulated environment.

Insulet Corporation (FDA-regulated medical devices) — Sr. Cloud Engineering Manager · Dec 2022 – Feb 2025

- Managed cloud and data-product teams (25–30 engineers) across DevOps, developer-productivity tooling, and performance engineering for an FDA-regulated connected medical-device platform.
- Owned end-to-end CI/CD (Jenkins, Python, Gradle) with pipeline gating; operated highly available Jenkins on EKS/EFS (99.9% uptime) and cut deployment errors 15%.
- **Performance engineering:** Built a performance and load-testing framework (Locust, Python) on EKS/AWS, running load and stress tests that drove a 25% improvement in cloud architecture performance.
- Implemented SBOM generation across ~25 microservices and a Python document-automation framework for regulated change control (ECO, SBR, PCO) — data-integrity and audit-trail rigor in a regulated SDLC.

Amazon (AWS Redshift) — Lead Systems Engineer · Dec 2016 – Dec 2022

- Owned the build, CI/CD, release ecosystem, and test tooling for Redshift — one of AWS's largest-scale distributed data services — including the CI/CD servers and fully automated build/deploy across all AWS-supported regions.
- Led the team that designed and shipped a Bors-style automated merge queue, gating integration on the exact merge candidate (not just the feature branch) — eliminating broken-mainline incidents and unblocking developer velocity org-wide.

- Added metrics-driven reporting on pipeline health and developer productivity, making bottlenecks visible and driving continuous improvement across the developer workflow.

Prosper Marketplace (regulated fintech) — Lead DevOps Engineer · Jan 2015 – Dec 2016

- Rolled out org-wide CI/CD with Jenkins and containerization with Docker; ran SonarQube static-analysis quality gates and Artifactory versioning on AWS.

Pivotal Software — Member of Technical Staff · Sep 2011 – Jan 2015

- Managed CI infrastructure; automated Coverity static analysis and code-coverage aggregation; built dynamic provisioning of ephemeral Linux test environments on AWS and vCloud.

OPEN SOURCE

UniClaw · github.com/uniclaw-labs/uniclaw

- Architected and shipped UniClaw, an open-source AI agent platform for connected vehicle control — a multi-provider framework supporting 27+ vehicle brands (Tesla, BMW, Ford, Rivian) with natural language commands via 12 messaging channels (WhatsApp, Telegram, Slack) and voice assistants (Alexa, Google Home), built on Claude-powered agents, a FastAPI (Python) backend, and a Tauri desktop app in React and TypeScript.
- Designed a security-first architecture: an OAuth2 authorization server with scoped JWTs, rate limiting, automatic Fernet encryption of secrets, constant-time token validation, and non-root containers.
- Delivered production-grade quality across a polyglot codebase (Python, TypeScript, Rust): 4,400+ automated tests, pluggable provider/channel adapters, and Dockerized multi-service orchestration.

EDUCATION

- M.S., Software Engineering, San Jose State University (2011)
- B.E., Computer Engineering, Gujarat University (2009)

SKILLS & TOOLS

- **Test Automation & Quality Engineering:** Test-automation platforms (Robot Framework); integration, E2E, regression & performance testing (Locust); pipeline gating, fixtures & reporting; AI-driven test generation (Gemini/LLM from specs); OPC UA / instrument-driven system testing; static analysis (SonarQube, Coverity)
- **CI/CD & Build Systems:** GitHub Actions, Jenkins, Bazel (monorepo, hermetic), Gradle; test orchestration & pipeline gating; self-service ephemeral environments; SBOM / secure pipelines
- **AI & Agents:** Multi-agent orchestration (miniBOT); RAG (Vertex AI, BigQuery embeddings); MCP servers (incl. published open-source); enterprise LLM rollout (Gemini Enterprise); AI-assisted development (Cursor, Claude Code)
- **Metrics & Reliability:** Quality SLOs; DORA metrics; observability; 99.99% reliability; SRE practices
- **Regulated & Data Integrity:** FDA-regulated medical-device software; regulated change control (ECO/SBR/PCO) & audit trails; ISO/IEC 27001 & SOC 2 (audit ownership)
- **Leadership & Strategy:** Building & leading small, high-leverage teams; hiring, mentoring & coaching; build-vs-buy, vendor & TCO; platform adoption through influence; roadmaps & OKRs
- **Software:** Python; TypeScript; Java/C/C#; Shell; API design